Weekly Electricity Consumption Matrix

1. Problem Statement

Develop a Python program to analyze electricity consumption for seven days using a matrix. The program should calculate the daily electricity consumption, appliance-wise total consumption, and identify days or appliances that exceed the specified electricity consumption limits.

2. Objectives

- To store electricity consumption data using a nested list (matrix).
- To calculate the total electricity consumption for each day.
- To calculate the total electricity consumption of each appliance over seven days.
- To compare consumption values with specified limits.
- To identify days and appliances with excessive electricity usage.

3. Assumptions

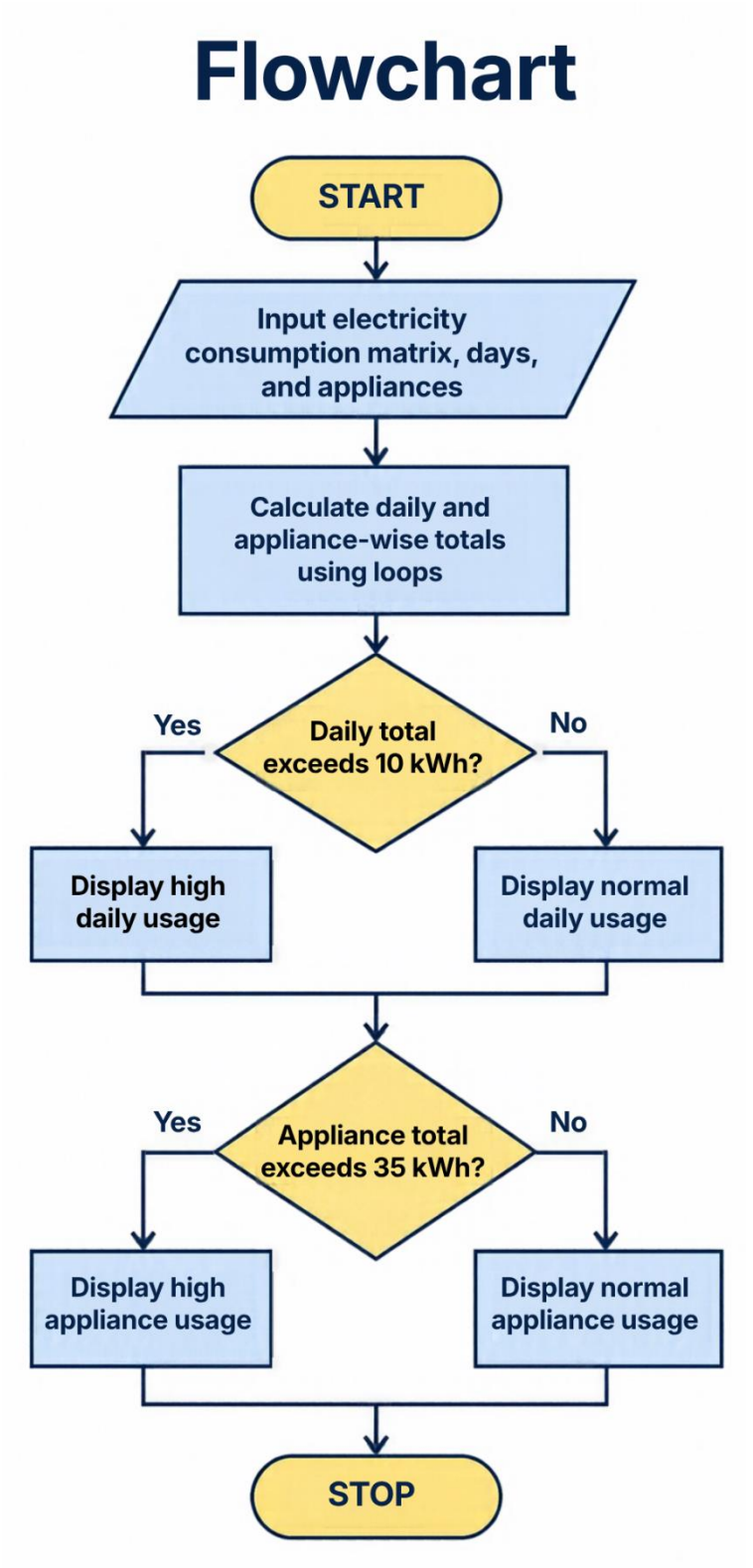
1. Electricity consumption is measured in kilowatt-hours (kWh).
2. The analysis is conducted for seven days, from Monday to Sunday.
3. Three appliances are considered: Fan, Refrigerator, and Air Conditioner.
4. Consumption values are entered as non-negative numbers.
5. The daily consumption limit is 10 kWh.
6. The weekly appliance consumption limit is 35 kWh.
7. The input data is assumed to be valid.

4. Algorithm

1. Start the program.
2. Define the names of seven days and three appliances.
3. Store electricity consumption values in a two-dimensional list.
4. Initialize the daily consumption limit and weekly appliance limit.

5. Calculate the total electricity consumption for each day using a loop.
6. Display each day's total and check whether it exceeds the daily limit.
7. Calculate the total consumption of each appliance over seven days.
8. Display each appliance's total and check whether it exceeds the weekly limit.
9. Display the results.
10. Stop the program.

5. Flowchart



6. Pseudocode

START

```
SET days = Monday to Sunday
SET appliances = Fan, Refrigerator, Air Conditioner

STORE electricity consumption in a matrix

SET daily_limit = 10
SET weekly_limit = 35

FOR each day:
    SET daily_total = 0
    FOR each appliance:
        ADD appliance consumption to daily_total
    DISPLAY daily_total

    IF daily_total > daily_limit:
        DISPLAY "Daily limit exceeded"

FOR each appliance:
    SET appliance_total = 0
    FOR each day:
        ADD daily consumption of appliance
    DISPLAY appliance_total
```

```
    IF appliance_total > weekly_limit:
        DISPLAY "Weekly appliance limit exceeded"

STOP
```

7. Python Actual Code

```
# Weekly Electricity Consumption Matrix

days = [
    "Monday", "Tuesday", "Wednesday",
    "Thursday", "Friday", "Saturday", "Sunday"
]

appliances = ["Fan", "Refrigerator", "Air Conditioner"]

# Electricity consumption in kWh
# Each row represents one day
# Each column represents one appliance

consumption = [
    [2, 3, 6],
    [2, 4, 5],
    [1, 3, 7],
    [2, 4, 6],
```

```
    [2, 3, 5],
    [3, 4, 7],
    [2, 3, 6]
]

daily_limit = 10
weekly_limit = 35

print("DAILY ELECTRICITY CONSUMPTION")

# Calculate daily totals
for i in range(len(days)):
    daily_total = 0

    for j in range(len(appliances)):
        daily_total += consumption[i][j]

    print(days[i], ":", daily_total, "kWh")

    if daily_total > daily_limit:
        print("Warning: Daily limit exceeded!")
    else:
        print("Daily consumption is within the limit.")
```

```
print("\nAPPLIANCE-WISE WEEKLY CONSUMPTION")

# Calculate appliance-wise totals
for j in range(len(appliances)):
    appliance_total = 0

    for i in range(len(days)):
        appliance_total += consumption[i][j]

    print(appliances[j], ":", appliance_total, "kWh")

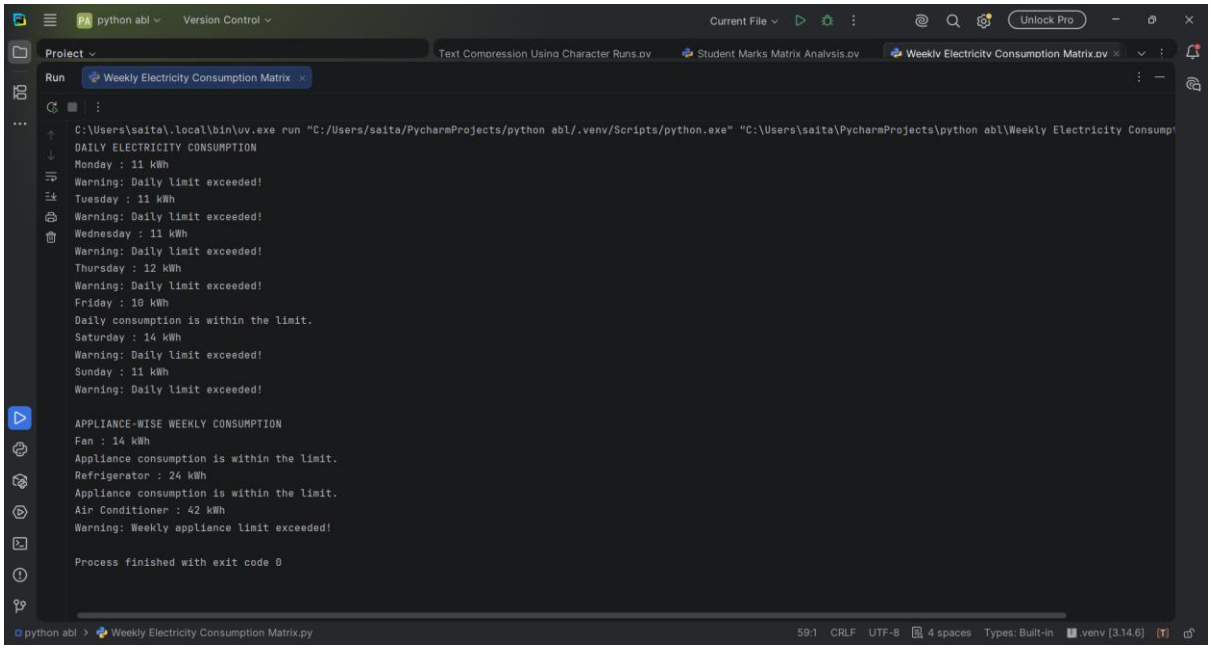
    if appliance_total > weekly_limit:
        print("Warning: Weekly appliance limit exceeded!")
    else:
        print("Appliance consumption is within the limit.")
```

8. Explanation of the Code

1. Days and appliances: Two lists store the seven days and the three appliance names.
2. Consumption matrix: A nested list stores daily electricity usage. Each row represents a day, and each column represents an appliance.
3. Limits: daily_limit is set to 10 kWh, and weekly_limit is set to 35 kWh.
4. Daily totals: The outer loop selects a day, and the inner loop adds the consumption of all appliances for that day.
5. Daily comparison: The if condition checks whether the daily total exceeds 10 kWh.

6. Appliance-wise totals: Another nested loop adds each appliance's consumption across all seven days.
7. Weekly comparison: The program checks whether each appliance's weekly consumption exceeds 35 kWh.
8. Output: The print() statements display totals and warnings when a limit is exceeded.

9. Output



10. Conclusion

The Python program successfully analyzes weekly electricity consumption using a matrix and nested loops. It calculates daily and appliance-wise totals and compares them with predefined limits. The program identifies days with excessive electricity usage and appliances that exceed their weekly consumption limit. This helps users monitor electricity usage and encourages energy conservation.